

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	11 September 2026
Team ID	Individual Project
Project Name	ServiceNow Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture:

The application uses ServiceNow as the main platform for the employee laptop request process.

S.No	Component	Description	Technology
1	User Interface	Employee, manager and IT users interact with the application through ServiceNow forms and lists.	ServiceNow UI / Forms / Lists
2	Application Logic-1	Logic for creating, validating and managing laptop requests.	ServiceNow Flow Designer
3	Application Logic-2	Logic for manager approval or rejection of laptop requests.	ServiceNow Approval / Flow Designer
4	Application Logic-3	Logic for status updates and user notifications.	ServiceNow Flow Designer / Notifications
5	Database	Stores employee laptop request records and related fields.	ServiceNow Database / Tables
6	Cloud Database	Platform-hosted data storage for application records.	ServiceNow Cloud Platform
7	File Storage	Files or attachments related to requests can be stored with ServiceNow records.	ServiceNow Attachments
8	External API-1	No external API is required for the core laptop request process.	Not Used
9	External API-2	No external API is required for the core laptop request process.	Not Used
10	Machine Learning Model	Machine learning is not required for the current application.	Not Used
11	Infrastructure (Server / Cloud)	Application is deployed and executed on the ServiceNow cloud platform.	ServiceNow Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source framework is required for the current ServiceNow application.	ServiceNow Platform
2	Security	Role-based access controls and ACLs restrict access to permitted laptop request records and functions.	ServiceNow Roles & ACLs
3	Scalable Architecture	The application is built on the ServiceNow platform and can support additional users and request records as the organization grows.	ServiceNow Cloud Platform
4	Availability	The application is hosted on the ServiceNow cloud platform and is accessible to authorized users through the platform.	ServiceNow Cloud
5	Performance	The application uses ServiceNow tables, forms, lists and Flow Designer to process and track requests efficiently.	ServiceNow Platform / Flow Designer

References

C4 Model – Architecture guidance

IBM Cloud Architecture – Architecture reference

AWS Architecture – Architecture reference